# Team clustering results vs Team squad composition

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# Main question

Are team clusters and team squad composition related?

# Do different clusters give different minutes to different player types?

At k = 2,

- Patient teams give more minutes for short passing keepers and line holding center backs.
- Front-footed teams give more minutes for passing wingbacks, while patient teams give more minutes for deep full backs.
- While soft shares were quite even, in terms of hard shares, front-footed teams give more minutes to progressive passing midfielders, while patient teams give more minutes to deep pivot midfielders and advanced carrying midfielders.
- Patient teams give more minutes for wide conservative carrying midfielders, while front-footed teams give more minutes for wide playmakers.
- Patient teams give more minutes for poachers, while front-footed teams give more minutes to advanced forwards.

At k = 3,

- Similar trends at the back can be seen as k = 2.
- Central combiners give more minutes to progressive passers but fewer to deep pivot midfielders.
- Central combiners give more minutes to wide playmakers, while wide crossers give more minutes to advanced wingers.
- Wide crossers give fewer minutes to poachers, but more to advanced forwards and target forwards.

At k = 4,

- High pressers show their front-footedness in more minutes to long passing keepers, stopper center backs, and progressive passers.

# Do the same trends remain across leagues?

A difference counts when it is at least 5 percent and points the same way in at least 3 of the 4 leagues.

- At k = 2, the patient/front-foot squad differences hold across leagues: patient teams give more minutes to advanced carriers, line holding center backs, and short passing keepers, while fewer minutes to progressive passers and progressive passing wingbacks.
- The signature for patient group remains the same throughout different values of k.
- At k = 3, central combiners give more minutes to progressive passers rather than deep pivots and more minutes to wide playmakers, while wide crossers give fewer minutes to advanced carriers and poachers.
- At k = 4, high pressing teams have more long passing keepers and stopper center backs but fewer advanced carriers.

Therefore, the squad pattern is mostly consistent among patient and front-footed teams, while differences among front-footed styles were smaller but still consistent across different leagues.

# Caveats

- Team style is made up of players' own actions, so the link between squad and style is automatic.
- 10 clubs have no clustered wingers, and 2 clubs have no clustered center forwards, so they sit out.
- Midfield difference is only seen in hard shares rather than soft, as midfielders' similarity scores are spread evenly across three types.

# Appendix

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## A1. Squad mix by cluster, k = 2

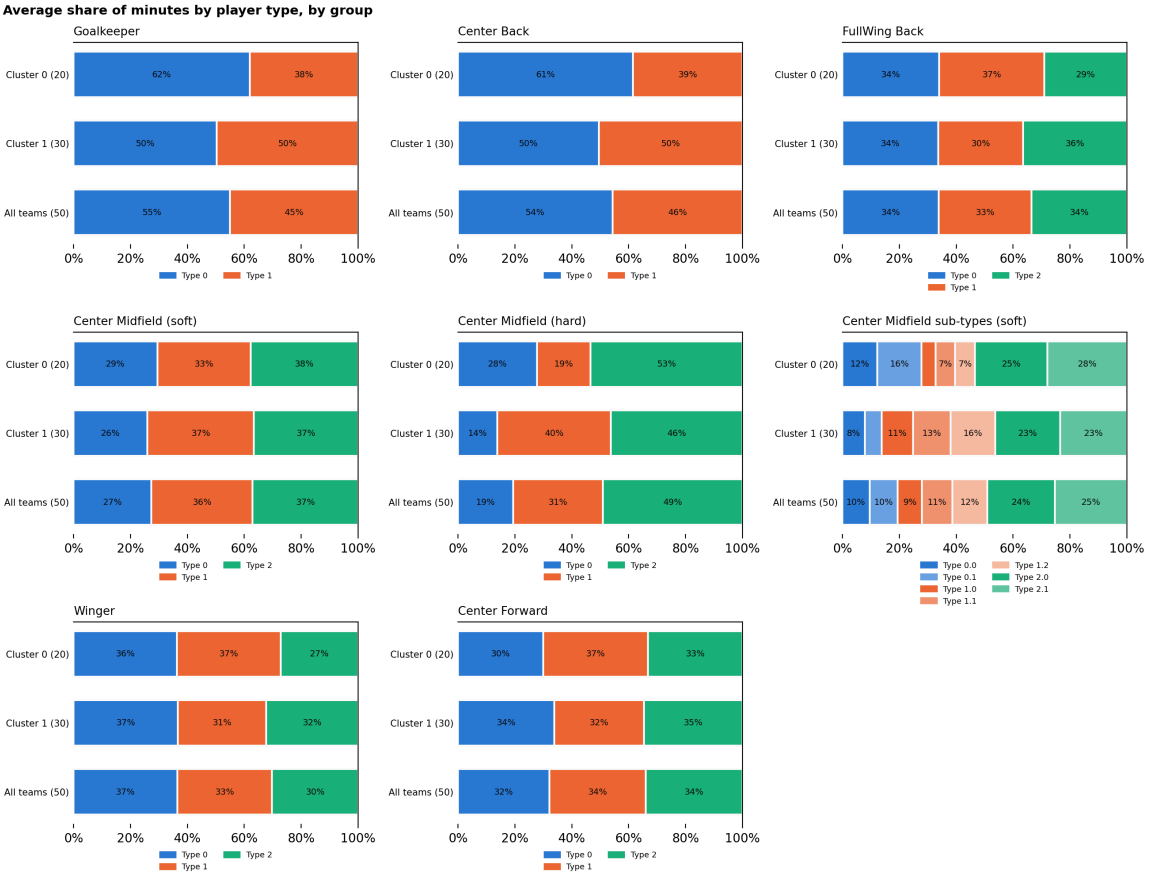



*Figure A1. Each cluster's share of minutes by player type at k = 2, against the 50-team average.*

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## A2. Player-type split by team, k = 2: goalkeeper and centre-back

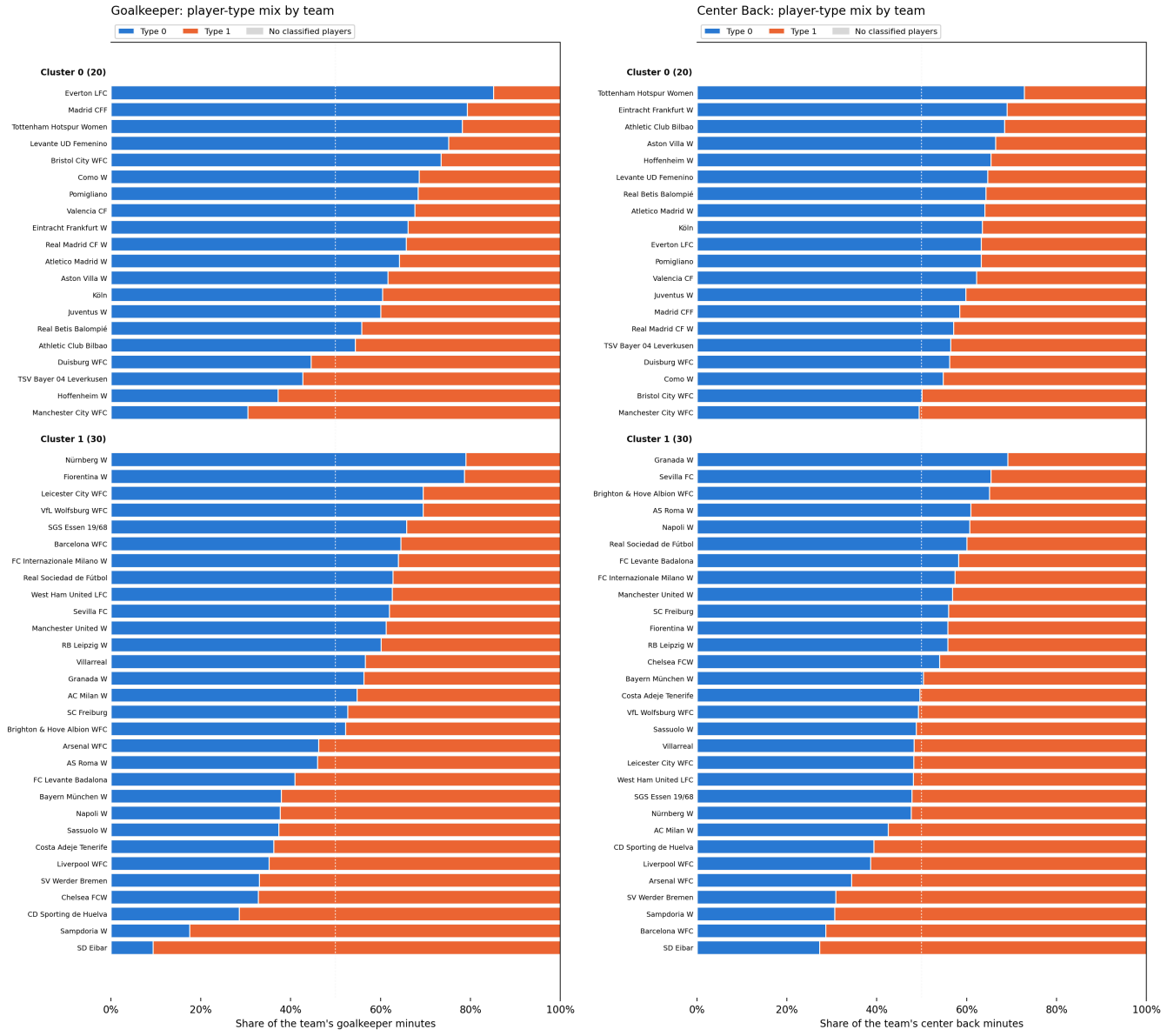



*Figure A2. Every team's split of goalkeeper (left) and centre-back (right) minutes across player types at k = 2, teams grouped by cluster.*

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## A3. Player-type split by team, k = 2: full/wing-back and midfield

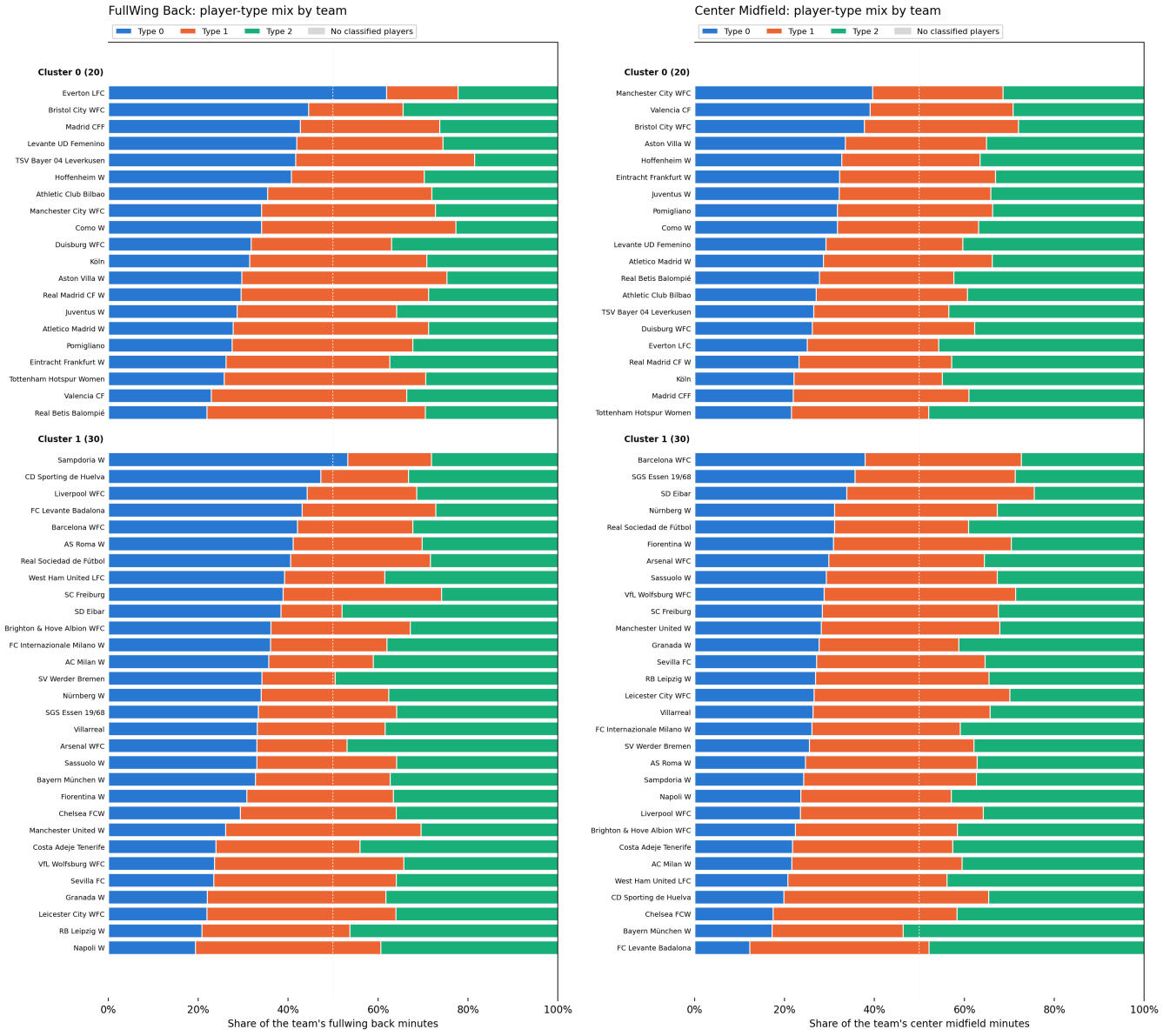



*Figure A3. Every team's split of full/wing-back (left) and midfield (right) minutes across player types at k = 2, teams grouped by cluster. Shares are soft shares, so the midfield difference seen in hard shares shows less here.*

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## A4. Midfield split by team, k = 2: hard types and sub-types

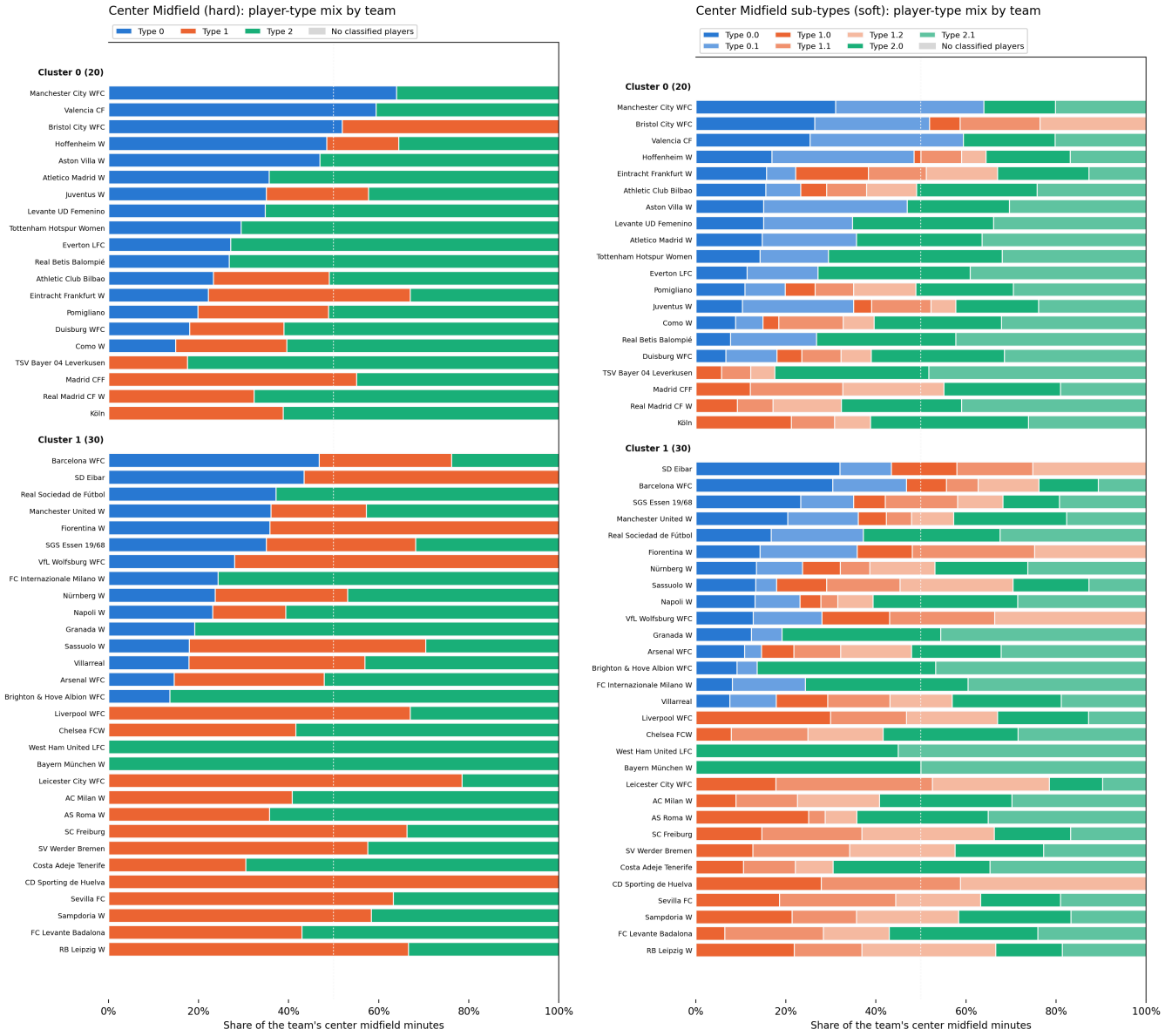



*Figure A4. Every team's split of midfield minutes at k = 2, teams grouped by cluster: hard types, with each midfielder counted as a single type (left), and midfield sub-types (right). This is where the midfield difference between styles shows.*

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## A5. Player-type split by team, k = 2: winger and centre-forward

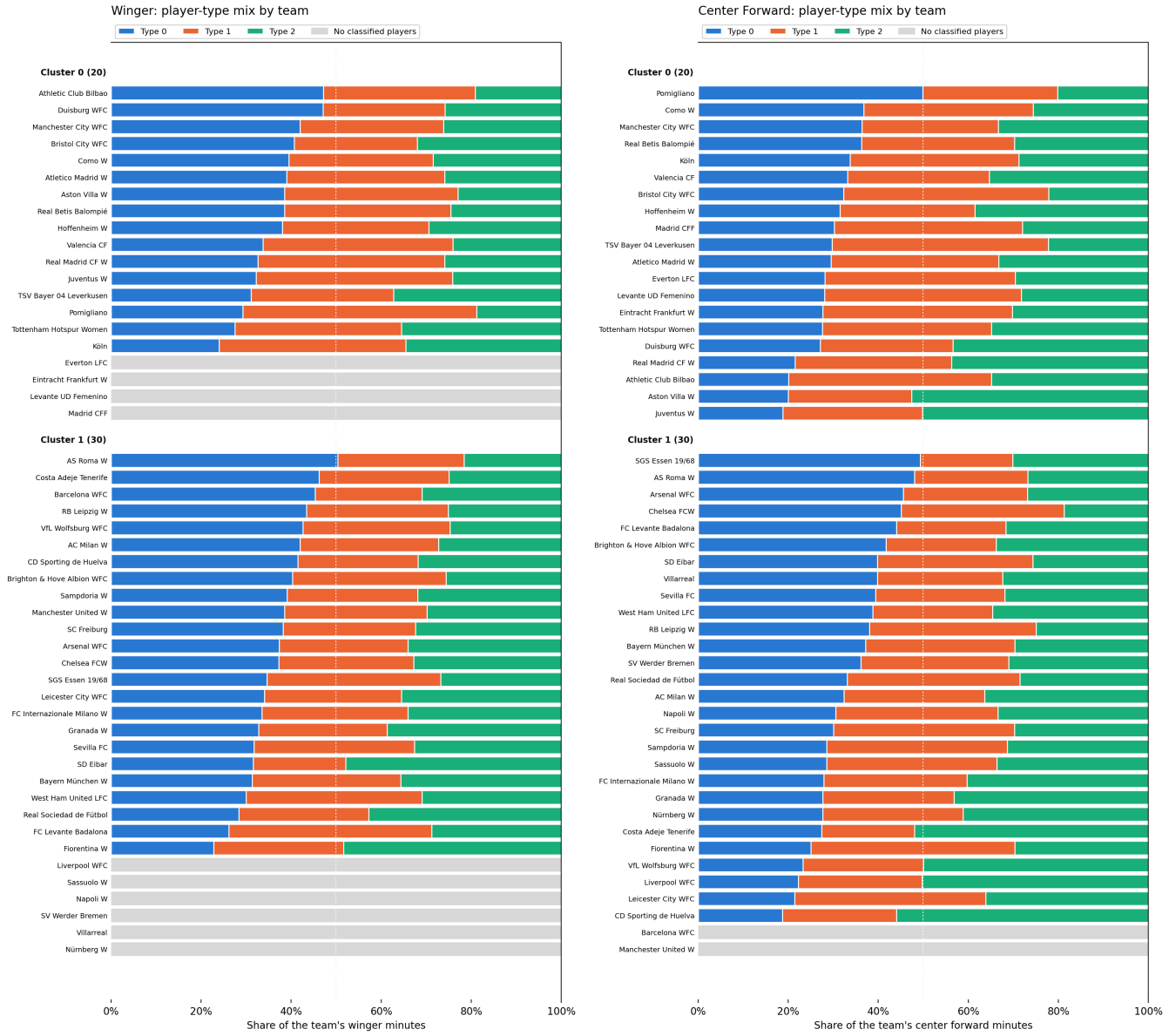



*Figure A5. Every team's split of winger (left) and centre-forward (right) minutes across player types at k = 2, teams grouped by cluster.*

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## A6. Squad mix by cluster, k = 3

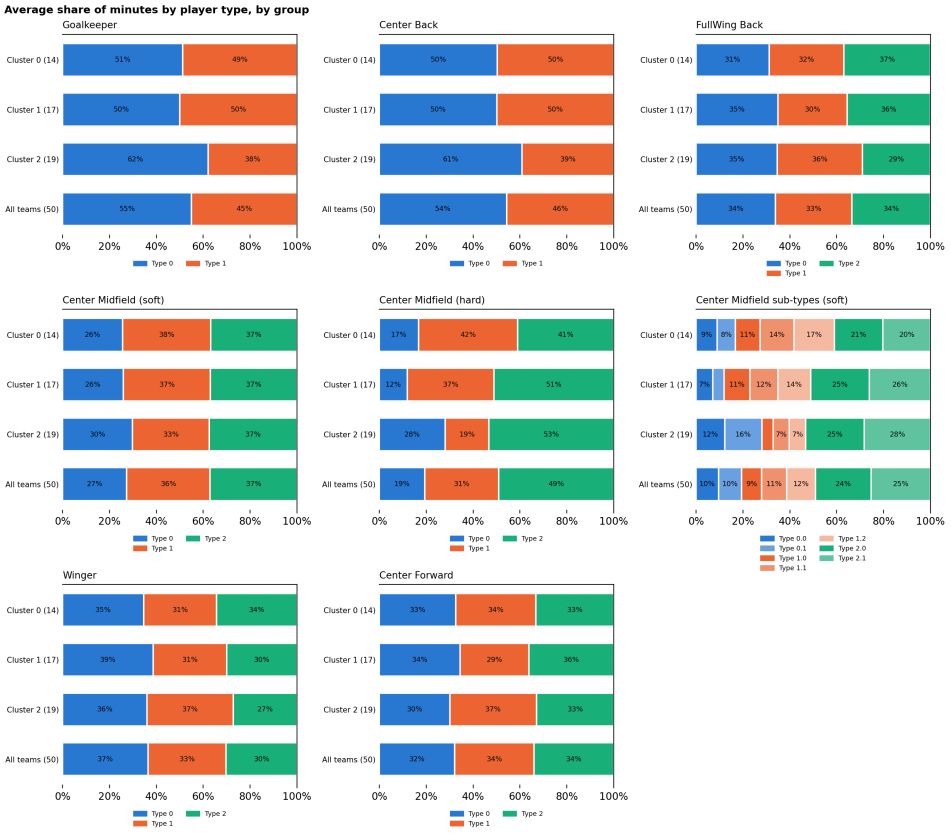



*Figure A6. Each cluster's share of minutes by player type at k = 3, against the 50-team average.*

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## A7. Player-type split by team, k = 3: goalkeeper and centre-back

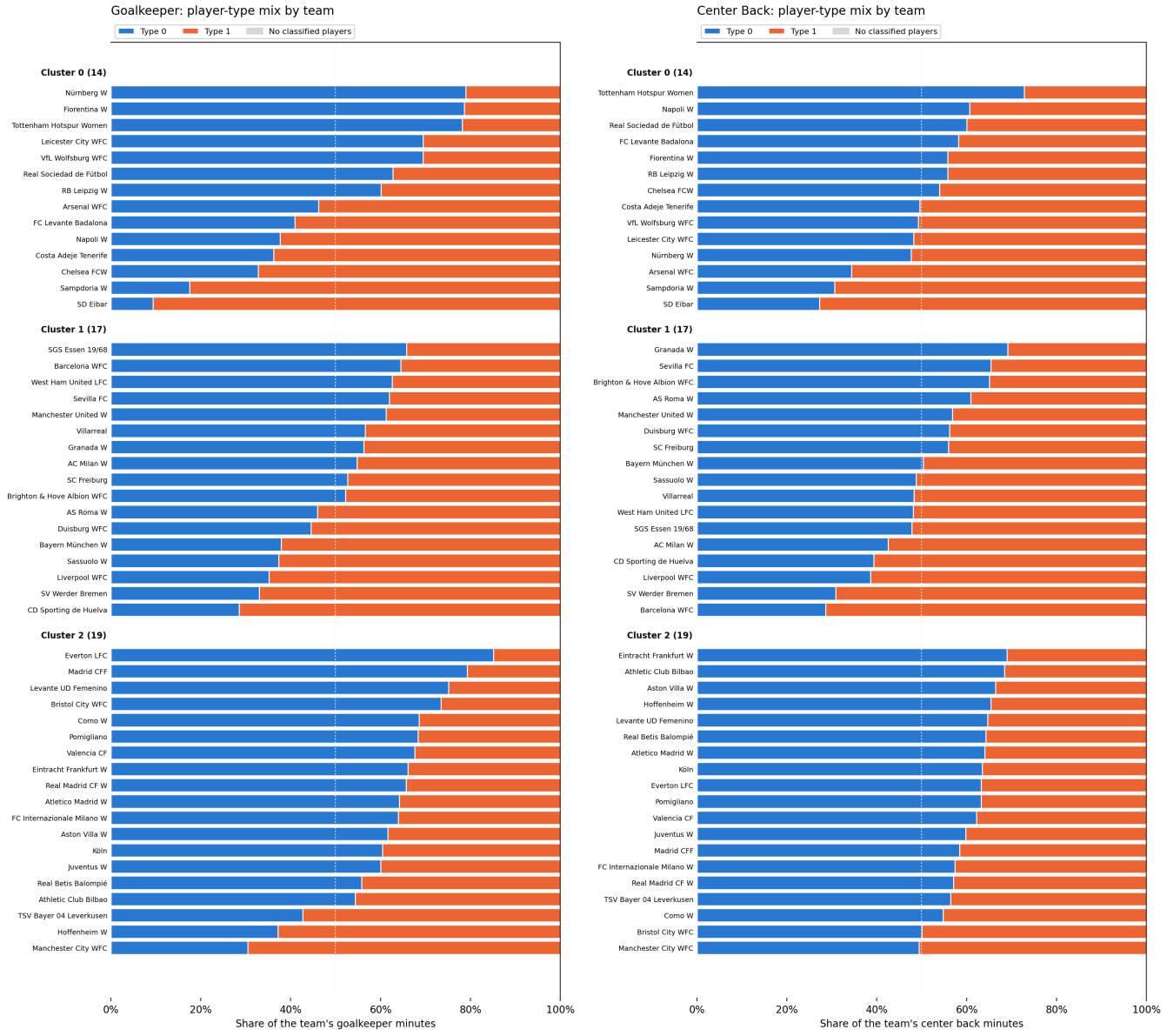



*Figure A7. Every team's split of goalkeeper (left) and centre-back (right) minutes across player types at k = 3, teams grouped by cluster.*

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## A8. Player-type split by team, k = 3: full/wing-back and midfield

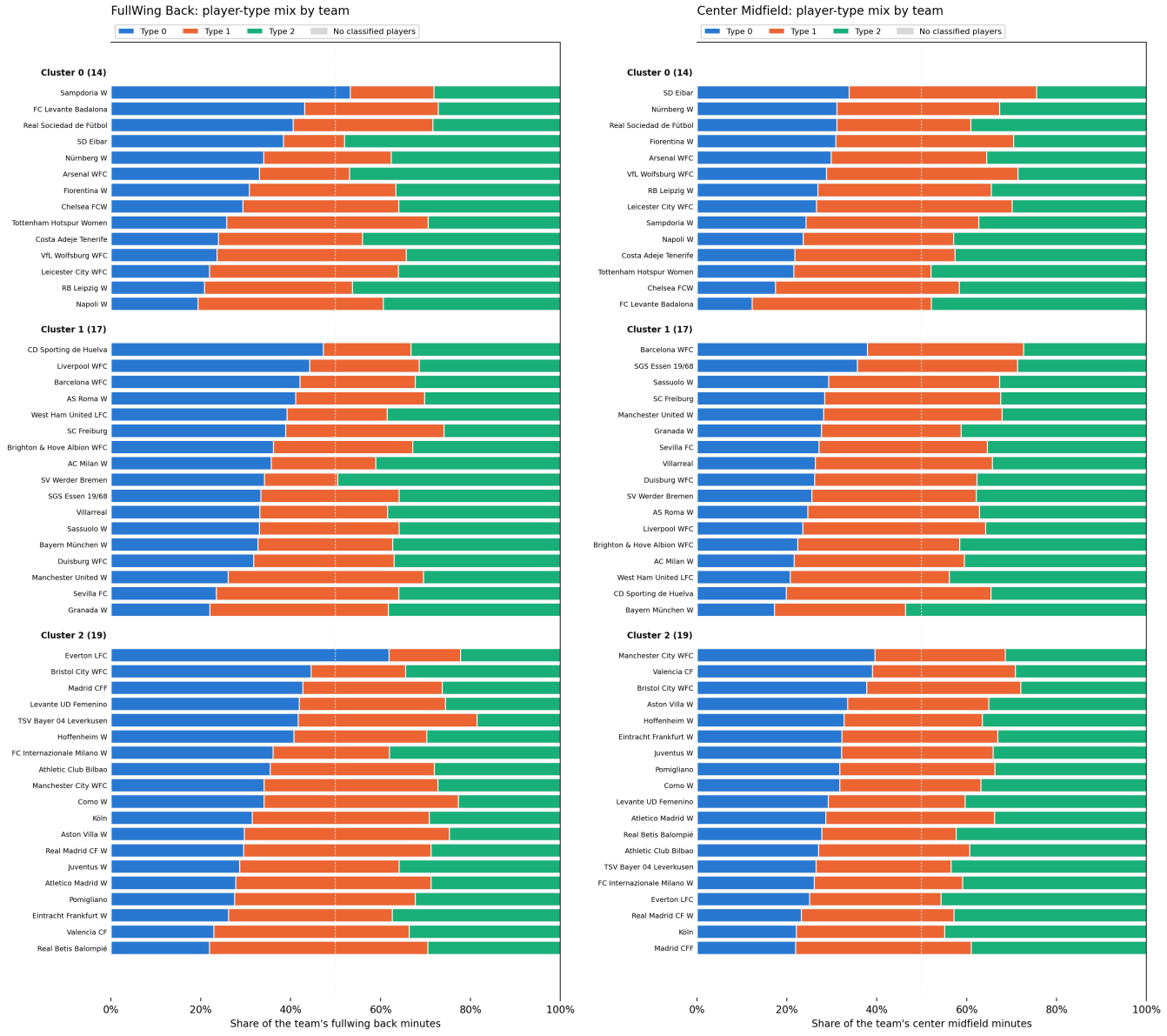



*Figure A8. Every team's split of full/wing-back (left) and midfield (right) minutes across player types at k = 3, teams grouped by cluster. Shares are soft shares, so the midfield difference seen in hard shares shows less here.*

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## A9. Midfield split by team, k = 3: hard types and sub-types

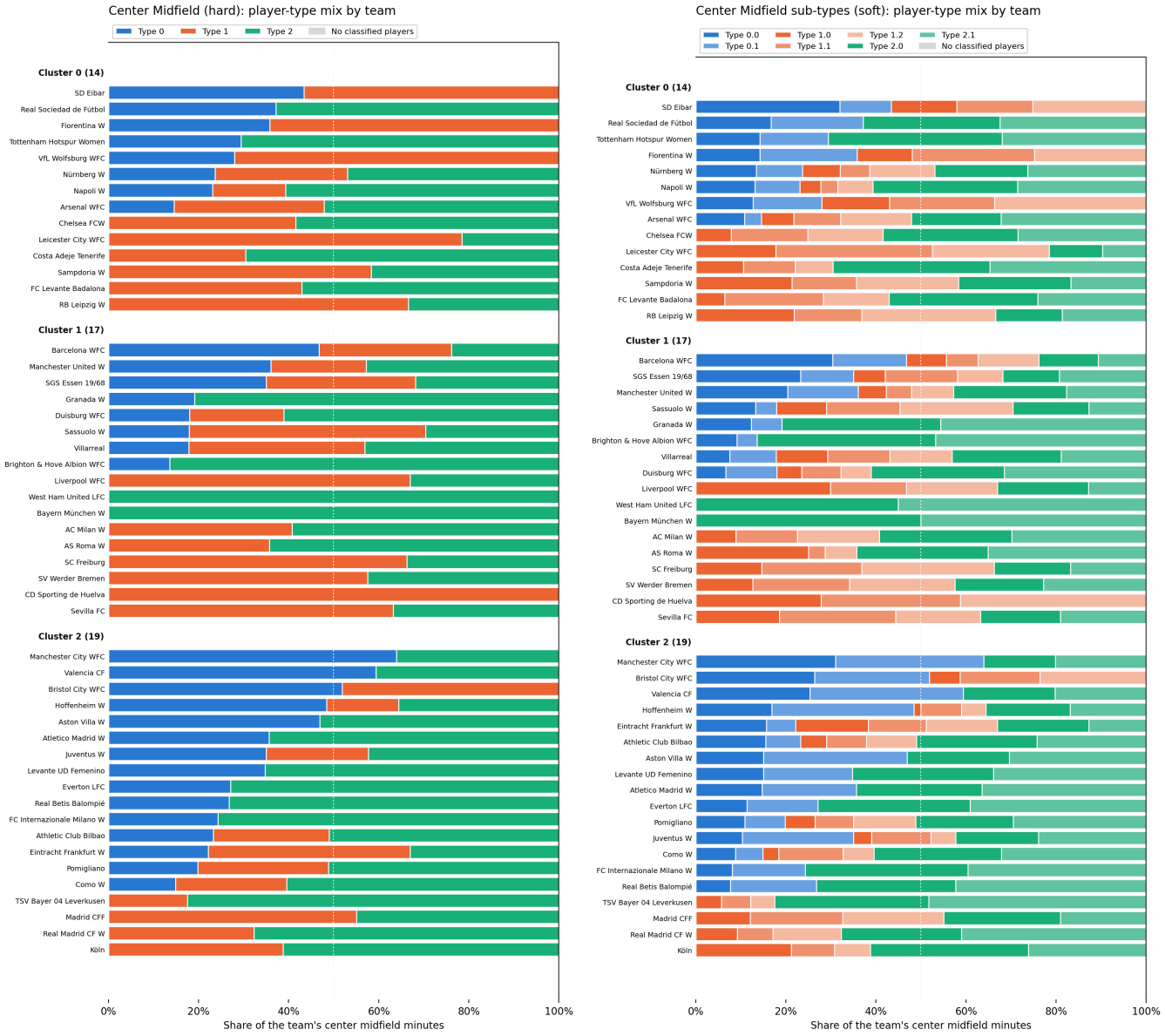



*Figure A9. Every team's split of midfield minutes at k = 3, teams grouped by cluster: hard types, with each midfielder counted as a single type (left), and midfield sub-types (right). This is where the midfield difference between styles shows.*

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## A10. Player-type split by team, k = 3: winger and centre-forward

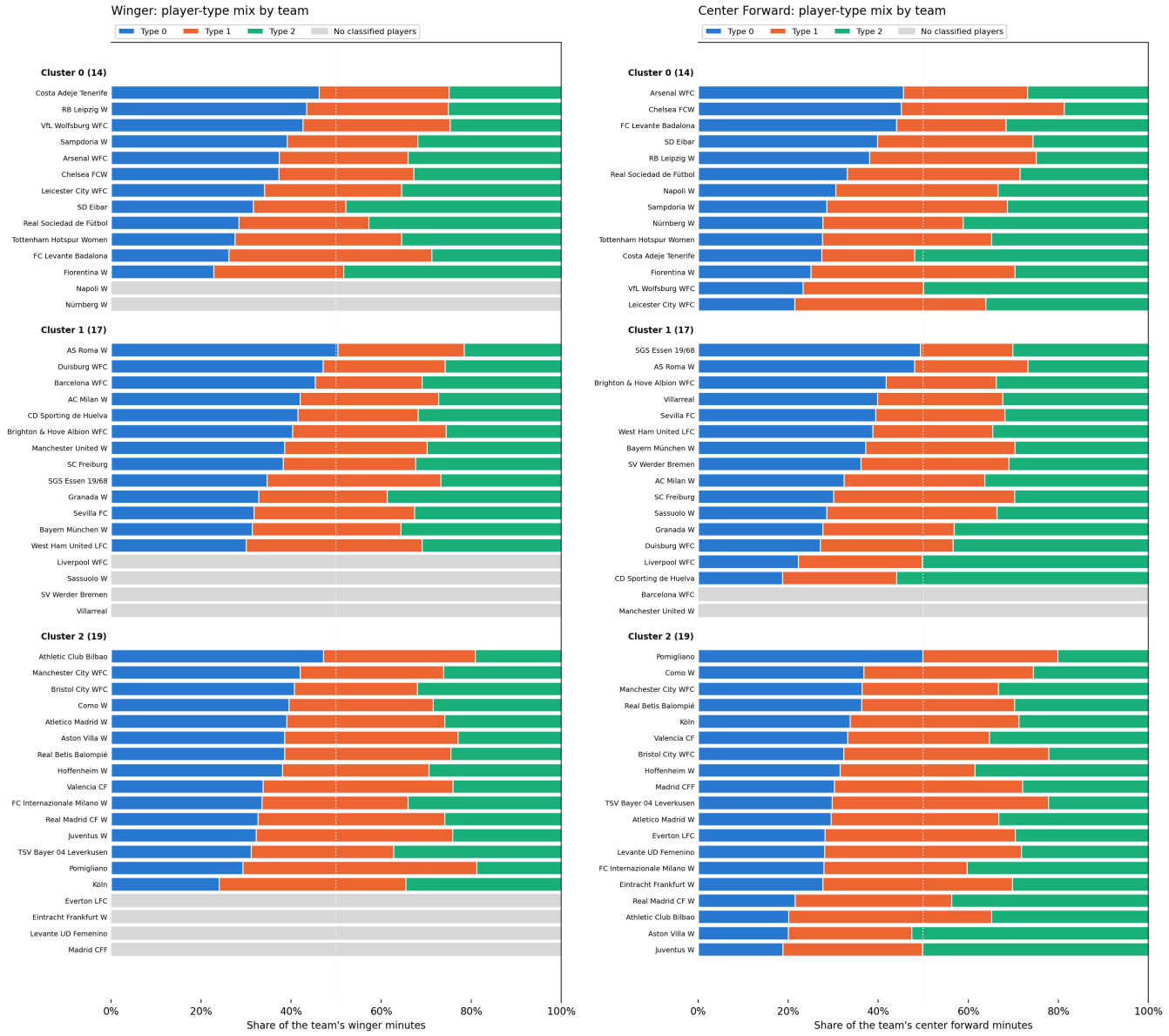



*Figure A10. Every team's split of winger (left) and centre-forward (right) minutes across player types at k = 3, teams grouped by cluster.*

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## A11. Squad mix by cluster, k = 4

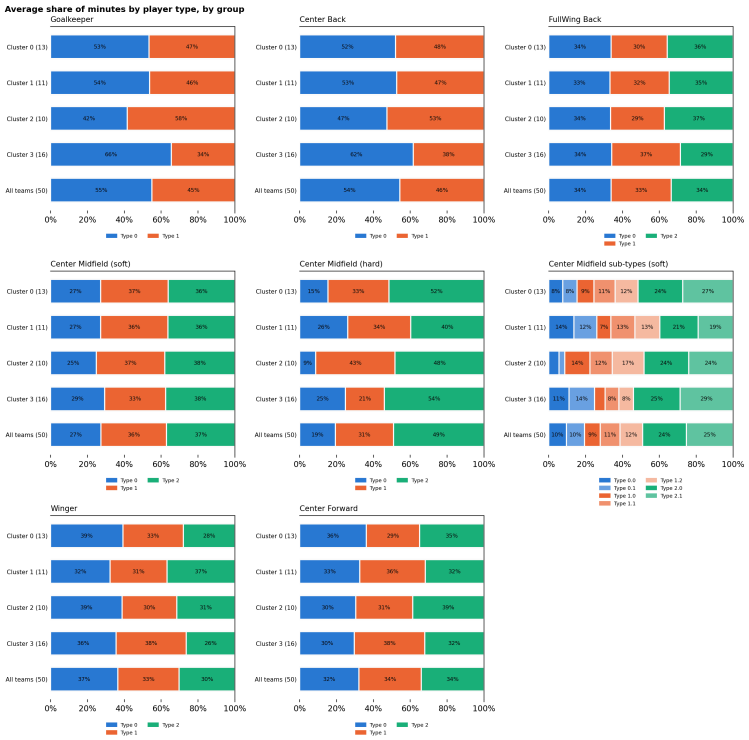



*Figure A11. Each cluster's share of minutes by player type at k = 4, against the 50-team average.*

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## A12. Player-type split by team, k = 4: goalkeeper and centre-back

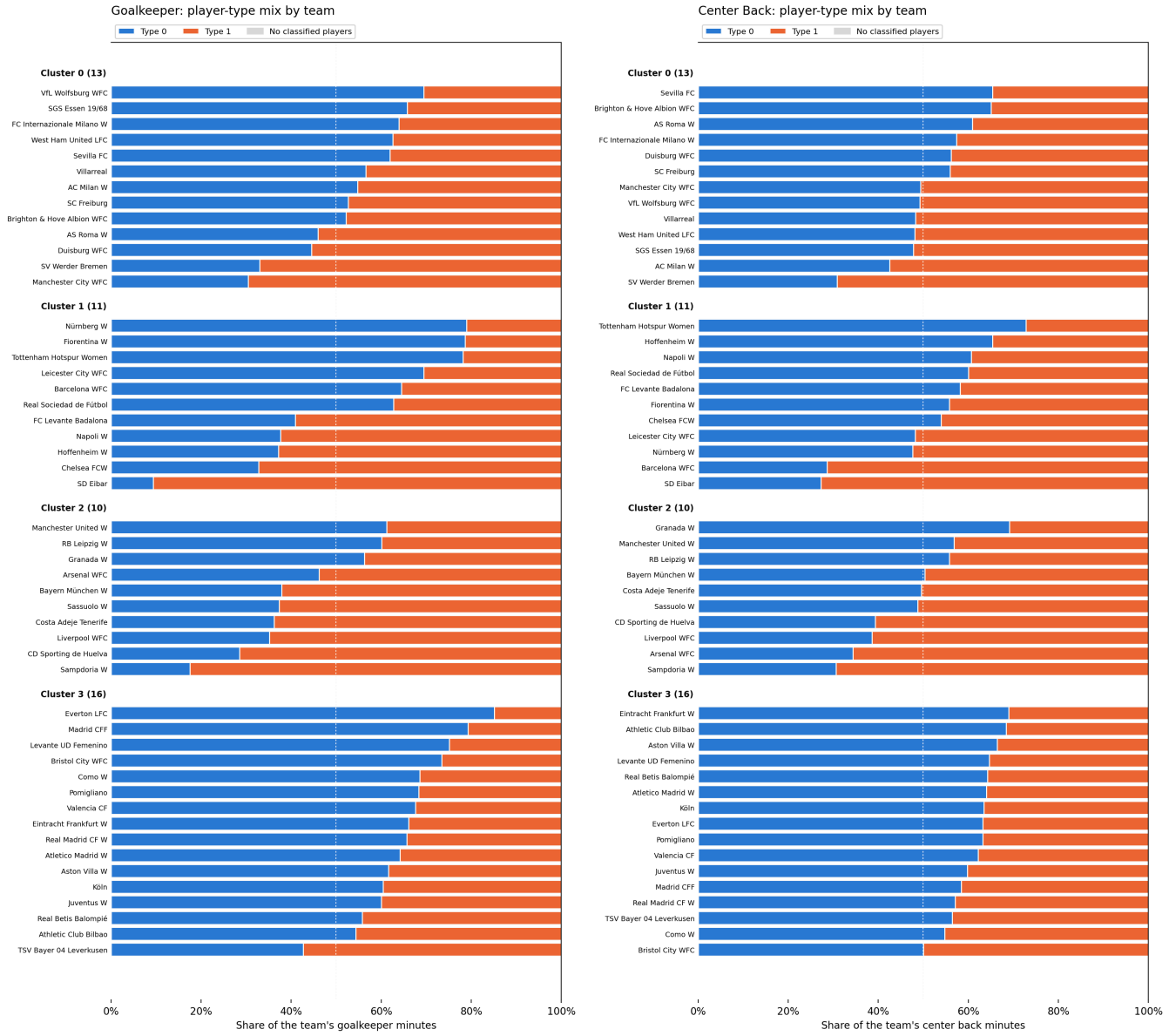



*Figure A12. Every team's split of goalkeeper (left) and centre-back (right) minutes across player types at k = 4, teams grouped by cluster.*

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## A13. Player-type split by team, k = 4: full/wing-back and midfield

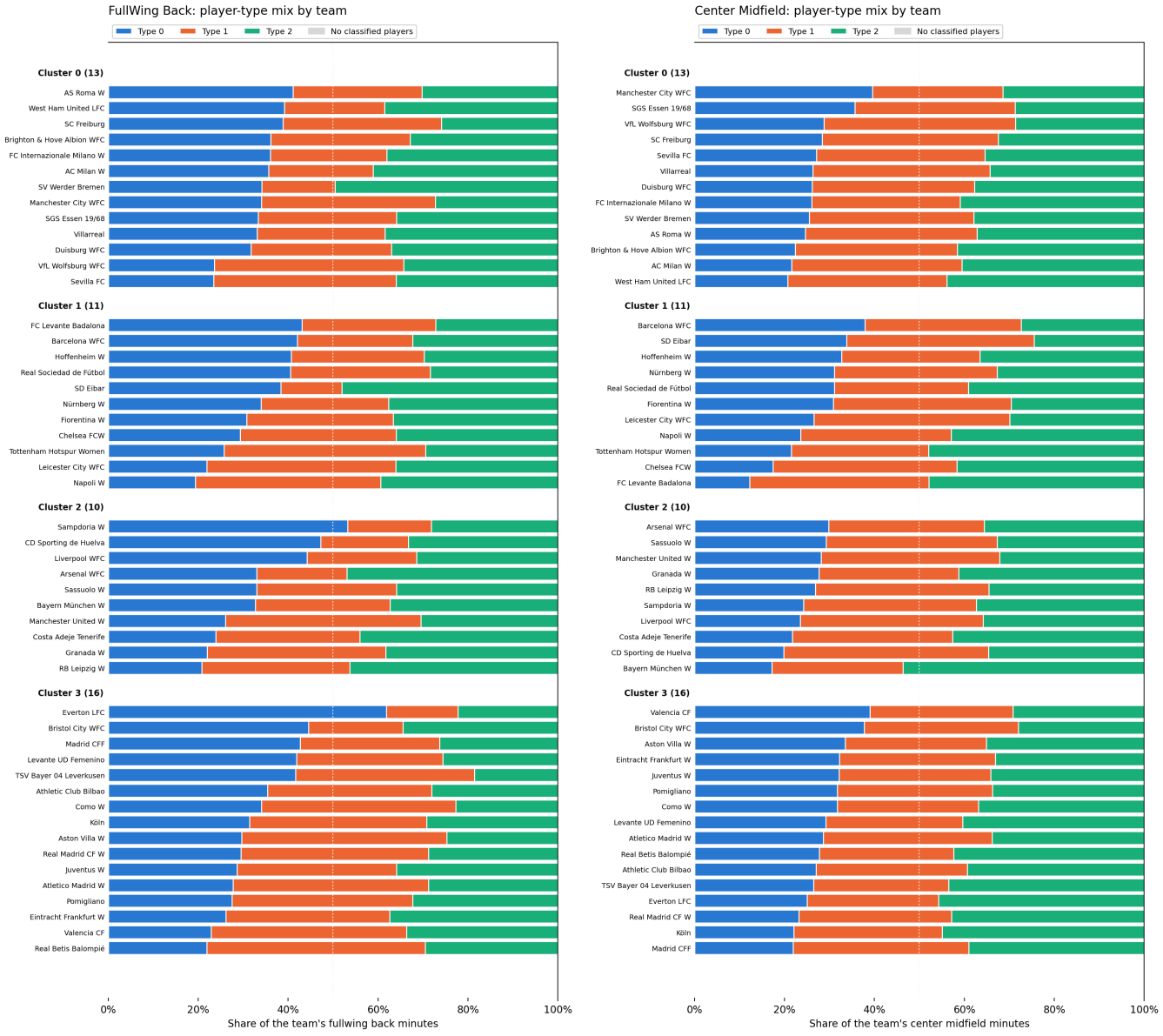



*Figure A13. Every team's split of full/wing-back (left) and midfield (right) minutes across player types at k = 4, teams grouped by cluster. Shares are soft shares, so the midfield difference seen in hard shares shows less here.*

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## A14. Midfield split by team, k = 4: hard types and sub-types

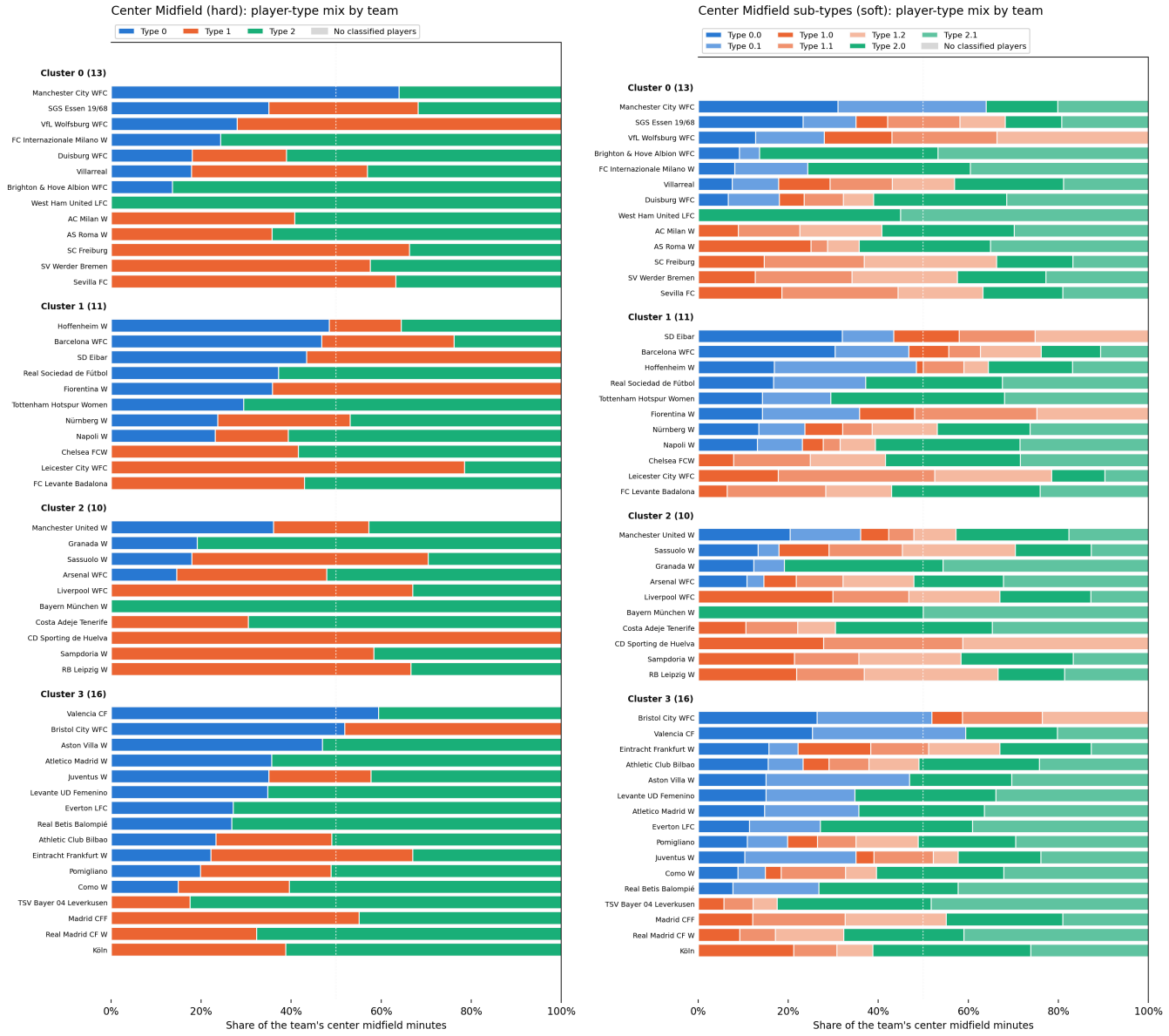



*Figure A14. Every team's split of midfield minutes at k = 4, teams grouped by cluster: hard types, with each midfielder counted as a single type (left), and midfield sub-types (right). This is where the midfield difference between styles shows.*

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## A15. Player-type split by team, k = 4: winger and centre-forward

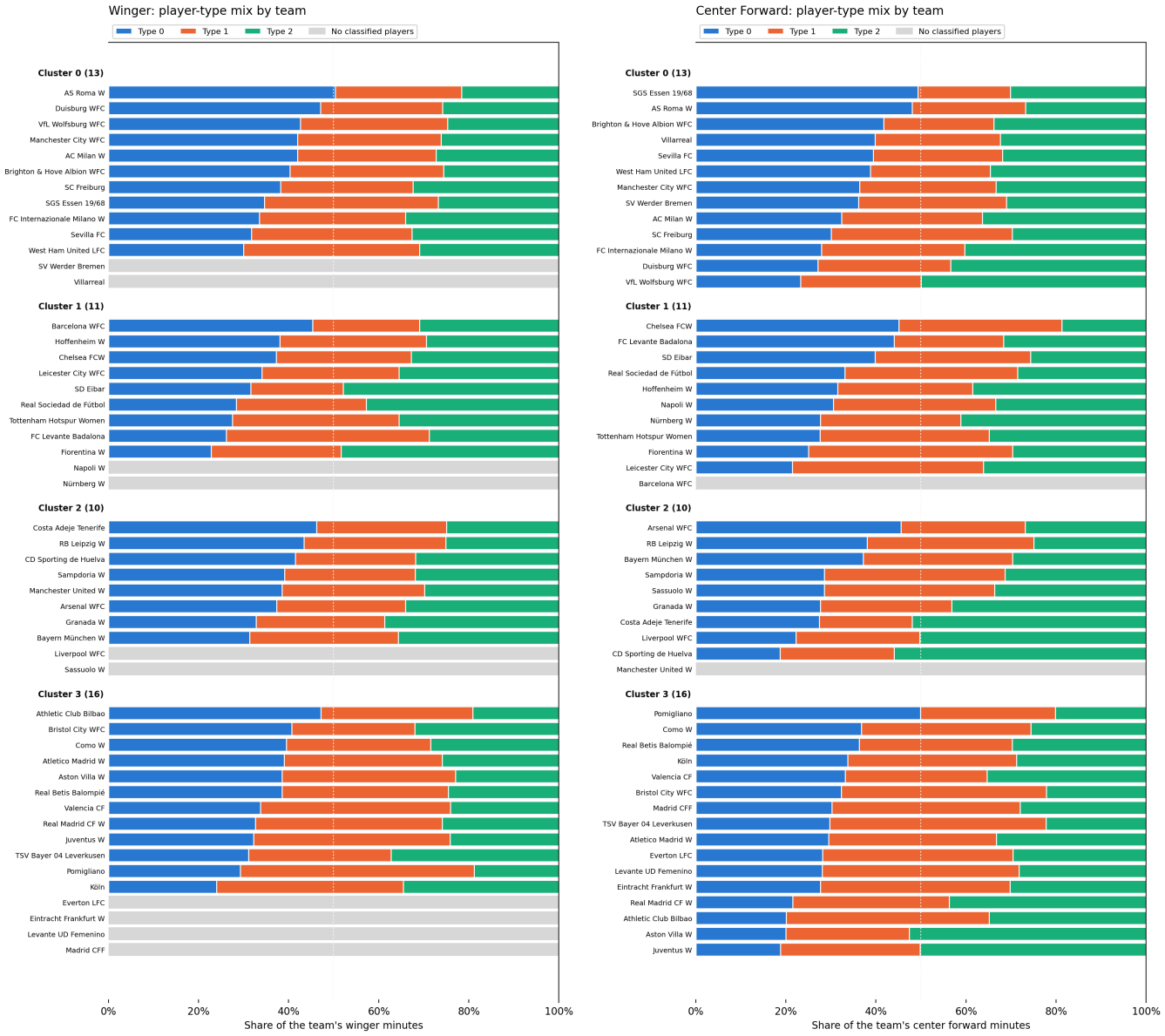



*Figure A15. Every team's split of winger (left) and centre-forward (right) minutes across player types at k = 4, teams grouped by cluster.*